MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3650 POX ACID WASH,ELTN. & CARBON REGN.
DESIGN OF TANK FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
ELTN.	:	Elution
REGN.	:	Regeneration

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1. INTENT

This document is to prepare design of foundation for **3650-THK-001,3650-REA-001, 3650-TNK- 002**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel and Staad. For member strength calculation force values are directly taken from Load Data Provided & Output of Staad analysis as applicable.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Input drawings attached in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 375 Kn/m^2 at 1.2 m below FGL. (Refer Table 5 - BH-11 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN OF FOUNDATION (3650-THK-001)

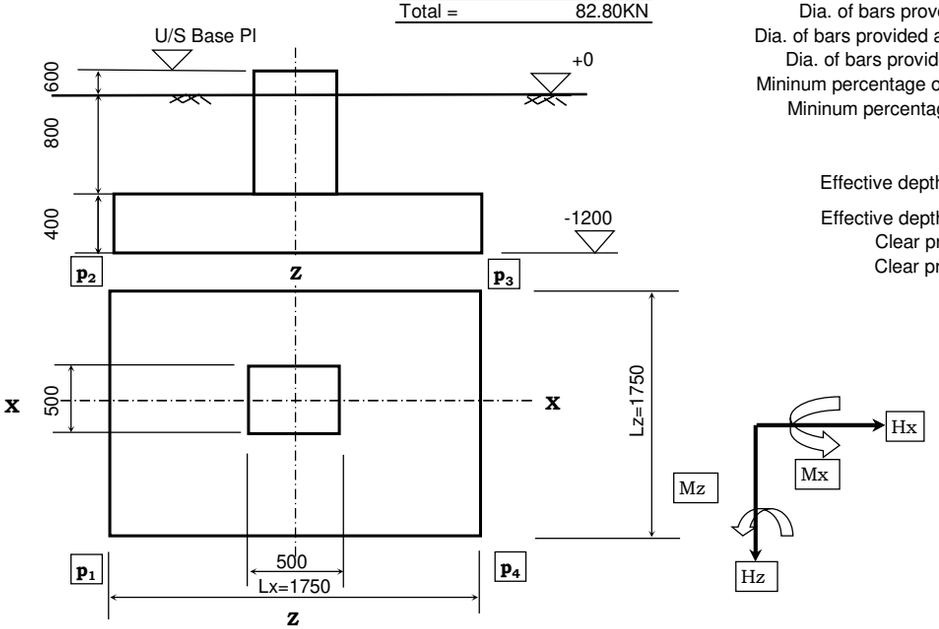
	Along-X	Along-Z
Size of column :	0.50m	0.50m
Size of fdn. raft :	1.75m	1.75m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.20 m	
U/S level Base Plate :	0.60 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	20 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	D
Seismic Load	E
Wind Load	W
Live Load	L

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.75*1.75*0.4*24)+(0.5*0.5*1.4*24) =	37.80KN
Weight of soil = ((1.75*1.75)-(0.5*0.5))*0.8*20 =	45.00KN
Total =	82.80KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 342mm = 0.342m

Effective depth of slab along Z direction = 326mm = 0.326m

Clear projection along Z direction = 0.625m

Clear projection along X direction = 0.625m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	14.00	0.00	0.00	0.00	0.00
2	Seismic Load (E)	25.00	11.00	0.00	0.00	0.00
3	Wind Load (W)	22.00	0.00	9.50	0.00	0.00
4	Live Load (L)	170.00	0.00	0.00	0.00	0.00

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D	14.0	0.0	0.0	0.0	0.0	96.80	0.00	0.00	0.00	0.00
202	D + L	184.0	0.0	0.0	0.0	0.0	266.80	0.00	0.00	0.00	0.00
203	D + W	36.0	0.0	0.0	9.5	0.0	118.80	0.00	0.00	9.50	17.10
204	0.6 D + W	30.4	0.0	0.0	9.5	0.0	113.20	0.00	0.00	9.50	17.10
205	0.6 D + 0.7 E	25.9	7.7	0.0	0.0	0.0	75.58	7.70	13.86	0.00	0.00
206	D + 0.75W + 0.75L	158.0	0.0	7.1	0.0	0.0	240.8	0.0	7.13	0.0	0.00
Ultimate Conditions											
301	1.4 D	19.6	0.0	0.0	0.0	0.0	135.52	0.00	0.00	0.00	0.00
302	1.2 D + 1.6 L	288.8	0.0	0.0	0.0	0.0	404.72	0.00	0.00	0.00	0.00
303	1.2 D + 1.0 E + 0.5 L	126.8	11.0	0.0	0.0	0.0	226.16	11.00	19.80	0.00	0.00
304	0.9 D + 1.0 E	37.6	11.0	0.0	0.0	0.0	112.12	11.00	19.80	0.00	0.00
305	0.9 D + 1.6 W	47.8	0.0	0.0	15.2	0.0	122.32	0.00	0.00	15.20	27.36
306	1.2 D + 1.6 W + 0.5 L	137.0	0.0	0.0	15.2	0.0	236.36	0.00	0.00	15.20	27.36

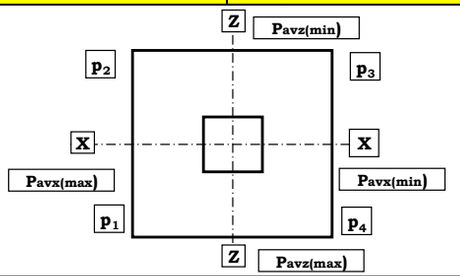
Check for overturning and sliding:

	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D	96.80	0.00	0.00	0.00	0.00	-	-	-	-
202	D + L	266.80	0.00	0.00	0.00	0.00	-	-	-	-
203	D + W	118.80	0.00	0.00	9.50	17.10	#DIV/0!	6.08	#DIV/0!	5.00
204	0.6 D + W	113.20	0.00	0.00	9.50	17.10	#DIV/0!	5.79	#DIV/0!	4.77
205	0.6 D + 0.7 E	75.58	7.70	13.86	0.00	0.00	4.77	#DIV/0!	3.93	#DIV/0!
206	D + 0.75W + 0.75L	240.80	0.00	7.13	0.00	0.00	29.57	-	#DIV/0!	-

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Load Combination & Calculation of Base pressure

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m²

Net SBC of soil for Dead Load = 375 kN/m² For Seismic load case = 468.75 kN/m²

Gross SBC of soil for Dead Load = 399 kN/m² For Seismic load case = 492.75 kN/m²

Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	D	96.80	0.00	0.00	31.61	0.00	0.00	31.61	31.61	31.61	31.61	7.61	31.61	31.61	31.61	31.61	O.K
202	D + L	266.80	0.00	0.00	87.12	0.00	0.00	87.12	87.12	87.12	87.12	63.12	87.12	87.12	87.12	87.12	O.K
203	D + W	118.80	0.00	17.10	38.79	0.00	19.14	57.94	19.65	57.94	19.65	33.94	38.80	38.80	57.94	19.65	O.K
204	0.6 D + W	113.20	0.00	17.10	36.96	0.00	19.14	56.11	17.82	56.11	17.82	32.11	36.97	36.97	56.11	17.82	O.K
205	0.6 D + 0.7 E	75.58	13.86	0.00	24.68	15.52	0.00	40.20	40.20	9.16	9.16	16.20	40.20	9.16	24.68	24.68	O.K
206	D + 0.75W + 0.75L	240.80	7.13	0.00	78.63	7.98	0.00	86.61	86.61	70.65	70.65	62.61	86.61	70.65	78.63	78.63	O.K
301	1.4 D	135.52	0.00	0.00	44.25	0.00	0.00	44.25	44.25	44.25	44.25	20.25	44.25	44.25	44.25	44.25	
302	1.2 D + 1.6 L	404.72	0.00	0.00	132.15	0.00	0.00	132.15	132.15	132.15	132.15	108.15	132.15	132.15	132.15	132.15	
303	1.2 D + 1.0 E + 0.5 L	226.16	19.80	0.00	73.85	22.17	0.00	96.01	96.01	51.68	51.68	72.01	96.01	51.68	73.85	73.85	
304	0.9 D + 1.0 E	112.12	19.80	0.00	36.61	22.17	0.00	58.78	58.78	14.44	14.44	34.78	58.78	14.44	36.61	36.61	
305	0.9 D + 1.6 W	122.32	0.00	27.36	39.94	0.00	30.63	70.57	9.31	70.57	9.31	46.57	39.94	39.94	70.57	9.31	
306	1.2 D + 1.6 W + 0.5 L	236.36	0.00	27.36	77.18	0.00	30.63	107.81	46.55	107.81	46.55	83.81	77.18	77.18	107.81	46.55	
Max :		405	20	27													
Min :		76	0	0													

Max pressure at base of foundation without surcharge excluding weight of soil (Dead load cases) = 7.61 kN/m2

Max pressure at base of foundation including surcharge excluding weight of soil (Dead load cases) = 27.61 kN/m2

Max pressure at base of foundation without surcharge excluding weight of soil (Dead+Live+Wind+Seismic load cases) = 108.15 kN/m2

Max pressure at base of foundation including surcharge excluding weight of soil (Dead+ Live+Wind+Seismic load cases) = 128.15 kN/m2

Sample calculation of values of base pressure:

Load case : D + L

Maximum pressure, p_1 (pmax) = 87.12+0.00+0.00= 87.12 kN/m²

p_2= 87.12+0.00-0.00= 87.12 kN/m²

p_3= 87.12-(0.00+0.00)= 87.12 kN/m²

p_4= 87.12-0.00+0.00= 87.12 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 87.12-1.2*20= 63.12 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (87.12+87.12)/2= 87.12 kN/m2

Minimum average Pressure along X, $p_{avx(minx)}$ = (87.12+87.12)/2= 87.12 kN/m2

Maximum average Pressure along Z, $p_{avz(max)}$ = (87.12+87.12)/2= 87.12 kN/m2

Minimum average Pressure along Z, $p_{avz(minx)}$ = (87.12+87.12)/2= 87.12 kN/m2

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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.5+2*0.5*0.326)+(0.5+2*0.5*0.326))*2

= 3.304 m

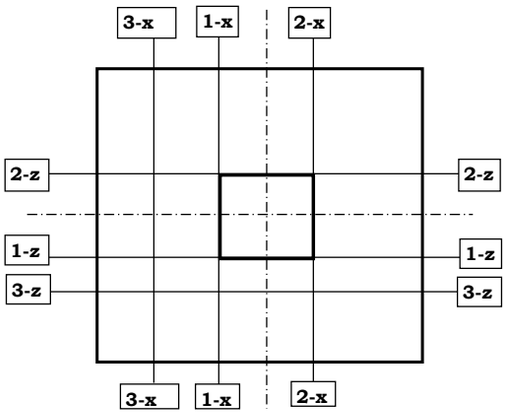
Punching Area= ((0.5+2*0.5*0.326)*(0.5+2*0.5*0.326))

= 0.68 m2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+0.8*20)

= 25.60 kN/m2

Area of footing, Af= 3.06 m2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.F(punching) kN
Ultimate Conditions								
301	1.4 D	44.25	44.25	44.25	3.64	3.64	5.28	44.44
302	1.2 D + 1.6 L	132.15	132.15	132.15	20.81	20.81	30.15	253.86
303	1.2 D + 1.0 E + 0.5 L	80.18	67.51	88.84	12.72	6.12	18.91	114.95
304	0.9 D + 1.0 E	42.94	30.28	51.61	5.45	-1.15	8.38	26.23
305	0.9 D + 1.6 W	39.94	39.94	39.94	2.80	2.80	4.06	34.17
306	1.2 D + 1.6 W + 0.5 L	77.18	77.18	77.18	10.07	10.07	14.60	122.89

Sample calculation of BM & SF

Load case : 301

p_{1x} = 44.25-(44.25-44.25)*0.625/1.75 = 44.25 kN/m^2

p_{2x} = 44.25+(44.25-44.25)*0.625/1.75 = 44.25 kN/m^2

p_{3x} = 44.25-(44.25-44.25)*(0.625-0.342)/1.75 = 44.25 kN/m^2

Overhang = 0.625 m

Effective depth = 0.342 m

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BM_{1x} = (44.25-25.6)*0.625*0.625/2+0.5*(44.25-44.25)*0.625*0.625*0.67 = 3.64 kN-m/m

BM_{2x} = (44.25-25.6)*0.625*0.625/2-0.5*(-44.25+44.25)*0.625*0.625*0.67 = 3.64 kN-m/m

SF_{3x} = (44.25-25.6)*(0.625-0.342)+0.5*(44.25-44.25)*(0.625-0.342) = 5.28 kN/m

$S.F_{(punching)}$ = (3.0625-0.68)*(44.25-25.6) = 44.44 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	20.81	0.18	0.24	0.001	193.95	615.60	16ϕT	326.61	300	200	1005.31
At top	-1.15	-0.01	0.01	0.000	10.67	205.20	10ϕT	382.75	300	200	392.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 30.15 kN/m

Shear stress , τ_v= (30.15*1000)/(1000*342)

= 0.09 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 253.86 kN

Strength of Concrete in shear Vc1 = 3342.49 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 2267.44 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 2228.33 kN Table 22.6.5.2_ (c)

Design Vc= 2228.33 kN

Allowable shear stress, $(\phi\ Vc)/bd$ = 1.48 N/mm2

Design punching shear stress, τ = (253.86*1000)/(3304*342)

= 0.22 N/mm2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
301	1.4 D	44.25	44.25	44.25	3.64	3.64	5.58
302	1.2 D + 1.6 L	132.15	132.15	132.15	20.81	20.81	31.86
303	1.2 D + 1.0 E + 0.5 L	73.85	73.85	73.85	9.42	9.42	14.43
304	0.9 D + 1.0 E	36.61	36.61	36.61	2.15	2.15	3.29
305	0.9 D + 1.6 W	48.69	31.19	60.10	7.37	-1.77	11.88
306	1.2 D + 1.6 W + 0.5 L	85.93	68.43	97.34	14.65	5.50	23.02

Sample calculation of BM & SF

Load case : 301

p_{1z} = 44.25-(44.25-44.25)*0.625/1.75= 44.25 kN/m2

p_{2z} = 44.25+(44.25-44.25)*0.625/1.75= 44.25 kN/m2 Overhang = 0.625 m

p_{3z} = 44.25-(44.25-44.25)*(0.625-0.326)/1.75= 44.25 kN/m2 Effective depth = 0.326 m

BM_{1z} = (44.25-25.6)*0.625*0.625/2+0.5*(44.25-44.25)*0.625*0.625*0.67 = 3.64 kN-m/m

BM_{2z} = (44.25-25.6)*0.625*0.625/2-0.5*(-44.25+44.25)*0.625*0.625*0.67 = 3.64 kN-m/m

SF_{3z} = (44.25-25.6)*(0.625-0.326)+0.5*(44.25-44.25)*(0.625-0.326) = 5.58 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm2)	ρ=	Ast (mm2)	Minm. Ast (mm2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm2)
At bottom	20.81	0.196	0.261	0.0006	213.54	586.80	16ϕT	342.64	300	200	1005.3
At top	-1.77	-0.017	-0.022	-0.0001	-0.17	195.60	10ϕT	401.53	300	200	392.7

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.74 N/mm^2

Shear at a distance 'd' from face of column, V_u= 31.86 kN/m

Shear stress , τ_v= (31.86*1000)/(1000*326)

= 0.10 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c
Along Z Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c

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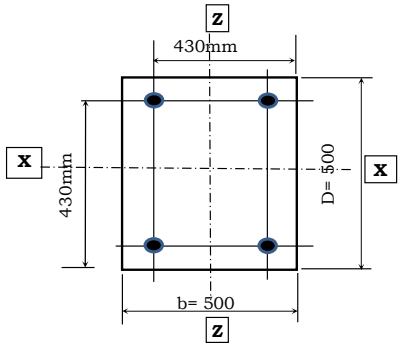
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DESIGN OF COLUMN FOR 3650-THK-001

Input:

Clear cover to reinf.=	50mm
Dia. of main reinf.=	20mm
Dia. of link=	10mm
Effective cover =	70mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%

Table 20.6.1.3.1 of SBC-304



For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	1400mm
Unsupproted length of column about Z (l_{uz}) =	1400mm

For Columns not braced against sidesway,

Where, Effective length factor K= 2

Slenderness ratio, $K l_{ux} / r_x$ = 18.67 < 22 SECTION 6.2.5a Radius of gyration r_x= 150 mm

$K l_{uz} / r_z$ = 18.67 < 22 SECTION 6.2.5a Radius of gyration r_z= 150 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	33.00	0.00	0.00	0.00	0.00
2	Seismic Load (E)	0.00	3.50	3.50	0.00	0.00
3	Wind Load (W)	0.00	5.00	5.00	0.00	0.00
4	Live Load (L)	170.00	0.00	0.00	0.00	0.00

γ_x =	0.76
γ_z =	0.76

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 D	4.62	0.00	0.00	7.11	0.00	0.00	0.01
302	1.2 D + 1.6 L	31.16	0.00	0.00	47.94	0.00	0.00	0.05
303	1.2 D + 1.0 E + 0.5 L	12.46	0.49	0.49	19.17	0.75	0.75	0.02
304	0.9 D + 1.0 E	2.97	0.49	0.49	4.57	0.75	0.75	0.01
305	0.9 D + 1.6 W	2.97	1.12	1.12	4.57	1.72	1.72	0.01
306	1.2 D + 1.6 W + 0.5 L	12.46	1.12	1.12	19.17	1.72	1.72	0.02

Reinforcement: Provide,

4-Nos	20 mm Dia	1257.14 Sq mm (At Corner)	Asc pvd. = 3771.43 Sq mm
4-Nos	20 mm Dia	1257.14 Sq mm (Along Z axis)	
4-Nos	20 mm Dia	1257.14 Sq mm (Along Y axis)	Total Asc = 1.5 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 890.93 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
301	0.05	0.05	21.9	21.9	0.55	1.16	0.00	O.K
302	0.05	0.05	21.9	21.9	0.55	1.16	0.00	O.K
303	0.05	0.05	21.9	21.9	0.55	1.16	0.04	O.K
304	0.05	0.05	21.9	21.9	0.55	1.16	0.04	O.K
305	0.05	0.05	21.9	21.9	0.55	1.16	0.11	O.K
306	0.05	0.05	21.9	21.9	0.55	1.16	0.11	O.K

Links, 10mm dia @ 300 C/C

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DESIGN CALCULATION OF FOUNDARION FOR 3650-REA-001

Size of Columns at node 8053,8055,8052 & 8054	Along-Z 0.50m	Along-X 0.50m
Size of fdn. raft:	2.25m	2.25m
Thickness of raft:	0.40 m	
Original Gr. Level:	0.00 m	
Finished Gr. Level:	0.00 m	
U/S of B/Pl Above GL:	0.55 m	
U/S level of fdn:	-1.20 m	
Unit weight of conc:	25 kN/m^3	
Unit weight of soil:	20 kN/m^3	

Calculation of Weight of Foundation :

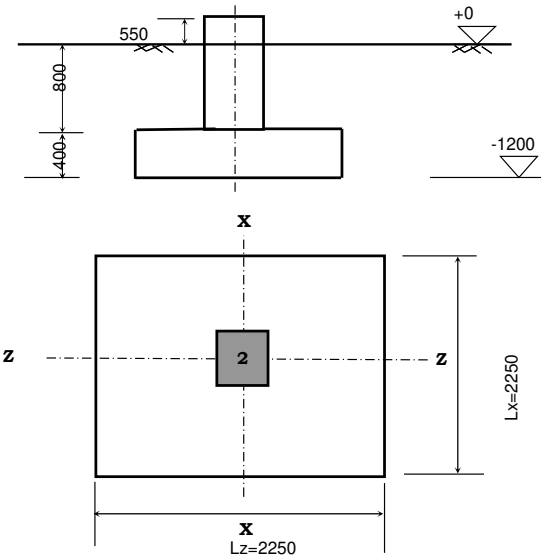
Weight of raft = (2.25*2.25*0.4*25) = 50.63KN

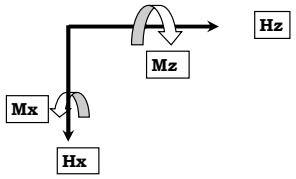
Weight of soil = ((2.25*2.25)-(0.5*0.5))*0.8*20 = 77.00KN

Total = 127.63KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding

Load at bottom of fooring	Node 8053					Node 8055				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + F	13.44	0.00	0.00	0.00	0.00	13.44	0.00	0.00	0.00	0.00
202 D + F + L	30.44	0.00	0.00	0.00	0.00	30.44	0.00	0.00	0.00	0.00
203 D + F + W	15.94	-2.00	3.50	-2.00	-3.50	15.94	-2.00	3.50	-2.00	-3.50
204 0.6D + W	10.56	-2.00	3.50	-2.00	-3.50	10.56	-2.00	3.50	-2.00	-3.50
205 0.6D + 0.7E	8.76	-0.70	1.23	-0.70	-1.23	8.76	-0.70	1.23	-0.70	-1.23
206 D + 0.75W + 0.75L	28.06	-1.50	2.63	-1.50	-2.63	28.06	-1.50	2.63	-1.50	-2.63

Load at bottom of fooring	Node 8052					Node 8054				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 D + F	13.44	0.00	0.00	0.00	0.00	13.438	0	0	0	0
202 D + F + L	30.44	0.00	0.00	0.00	0.00	30.438	0	0	0	0
203 D + F + W	15.94	-2.00	3.50	-2.00	-3.50	15.938	-2	3.5	-2	-3.5
204 0.6D + W	10.56	-2.00	3.50	-2.00	-3.50	10.563	-2	3.5	-2	-3.5
205 0.6D + 0.7E	8.76	-0.70	1.23	-0.70	-1.23	8.763	-0.7	1.225	-0.7	-1.225
206 D + 0.75W + 0.75L	28.06	-1.50	2.63	-1.50	-2.63	28.063	-1.5	2.625	-1.5	-2.625

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 D + F	181.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
202 D + F + L	249.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
203 D + F + W	191.4	-8.00	-14.0	-8.0	14.0	15.4	15.4	9.6	9.6
204 0.6D + W	169.9	-8.00	-14.0	-8.0	14.0	13.7	13.7	8.5	8.5
205 0.6D + 0.7E	162.7	-2.80	-4.9	-2.8	4.9	37.4	37.4	23.2	23.2
206 D + 0.75W + 0.75L	239.9	-6.00	-10.5	-6.0	10.5	25.7	25.7	16.0	16.0

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Base pressure check

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	7787	201	0	59.89	0
Min Px	7787	201	0	59.89	0
Max Py	8041	203	0	158.40	0
Min Py	8051	205	0	44.27	0
Max Pz	7787	201	0	59.89	0
Min Pz	7787	201	0	59.89	0

Gross Maximum Base pressure, p_{max} = 158.4 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.8 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 366 KN/m^2
> 158.4 KN/m^2
Hence OK

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Design of foundation raft

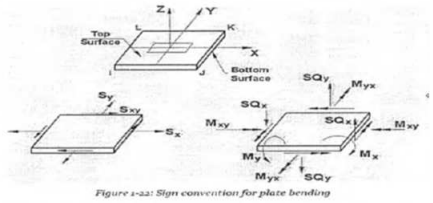
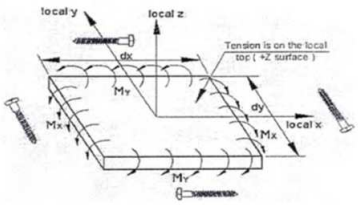


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	415	0.065	-0.059	0	0	0	0.807	0.982	0.26	1.067	1.242	102:1.2D + 1.2F + 1.6L
Min Qx	374	-0.085	-0.077	0	0	0	5.262	4.872	2.502	7.764	7.374	106:1.2D + 1.6W + 0.5L
Max Qy	421	0.061	0.063	0	0	0	0.926	0.864	-0.26	1.186	1.124	102:1.2D + 1.2F + 1.6L
Min Qy	374	-0.085	-0.077	0	0	0	5.262	4.872	2.502	7.764	7.374	106:1.2D + 1.6W + 0.5L
Max Sx	425	-0.01	-0.011	0	0	0	1.01	0.958	-0.096	1.106	1.054	101:1.4D + 1.4F
Min Sx	425	-0.01	-0.011	0	0	0	1.01	0.958	-0.096	1.106	1.054	101:1.4D + 1.4F
Max Sy	425	-0.01	-0.011	0	0	0	1.01	0.958	-0.096	1.106	1.054	101:1.4D + 1.4F
Min Sy	425	-0.01	-0.011	0	0	0	1.01	0.958	-0.096	1.106	1.054	101:1.4D + 1.4F
Max Sxy	425	-0.01	-0.011	0	0	0	1.01	0.958	-0.096	1.106	1.054	101:1.4D + 1.4F
Min Sxy	425	-0.01	-0.011	0	0	0	1.01	0.958	-0.096	1.106	1.054	101:1.4D + 1.4F
Max Mx	374	-0.085	-0.077	0	0	0	5.262	4.872	2.502	7.764	7.374	106:1.2D + 1.6W + 0.5L
Min Mx	421	-0.016	-0.017	0	0	0	-4.078	-4.216	-2.789	-6.867	-7.005	105:0.9D + 1.6W
Max My	374	-0.085	-0.077	0	0	0	5.262	4.872	2.502	7.764	7.374	106:1.2D + 1.6W + 0.5L
Min My	421	-0.016	-0.017	0	0	0	-4.078	-4.216	-2.789	-6.867	-7.005	105:0.9D + 1.6W
Max Mxy	374	-0.066	-0.06	0	0	0	5.03	4.589	2.577	7.607	7.166	105:0.9D + 1.6W
Min Mxy	421	0.001	0.001	0	0	0	-3.81	-3.967	-2.864	-6.674	-6.831	106:1.2D + 1.6W + 0.5L

PLATE CENTER STRESS : FDN_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	282	0.016	-0.005	0	0	0	-3.205	1.038	-0.168	-3.373	1.206	102:1.2D + 1.2F + 1.6L
Min Qx	311	-0.033	0.008	0	0	0	-1.531	0.067	0.12	-1.651	0.187	106:1.2D + 1.6W + 0.5L
Max Qy	377	-0.004	0.016	0	0	0	0.991	-3.222	-0.186	1.177	-3.408	102:1.2D + 1.2F + 1.6L
Min Qy	418	0.007	-0.033	0	0	0	0.072	-1.545	0.125	0.197	-1.67	106:1.2D + 1.6W + 0.5L
Max Sx	420	0.002	0.007	0	0	0	0.429	-1.4	0.08	0.509	-1.48	101:1.4D + 1.4F
Min Sx	420	0.002	0.007	0	0	0	0.429	-1.4	0.08	0.509	-1.48	101:1.4D + 1.4F
Max Sy	420	0.002	0.007	0	0	0	0.429	-1.4	0.08	0.509	-1.48	101:1.4D + 1.4F
Min Sy	420	0.002	0.007	0	0	0	0.429	-1.4	0.08	0.509	-1.48	101:1.4D + 1.4F
Max Sxy	420	0.002	0.007	0	0	0	0.429	-1.4	0.08	0.509	-1.48	101:1.4D + 1.4F
Min Sxy	420	0.002	0.007	0	0	0	0.429	-1.4	0.08	0.509	-1.48	101:1.4D + 1.4F
Max Mx	375	0.003	-0.024	0	0	0	1.334	-0.432	0.251	1.585	-0.683	106:1.2D + 1.6W + 0.5L
Min Mx	312	-0.003	0.009	0	0	0	-3.811	-0.018	-0.021	-3.832	-0.039	106:1.2D + 1.6W + 0.5L
Max My	281	-0.024	0.003	0	0	0	-0.429	1.392	0.241	-0.67	1.633	106:1.2D + 1.6W + 0.5L
Min My	420	0.009	-0.003	0	0	0	-0.025	-3.827	-0.005	-0.03	-3.832	106:1.2D + 1.6W + 0.5L
Max Mxy	294	0.008	-0.008	0	0	0	-2.833	-2.822	0.377	-3.21	-3.199	102:1.2D + 1.2F + 1.6L
Min Mxy	293	-0.008	-0.008	0	0	0	-2.838	-2.821	-0.377	-3.215	-3.198	102:1.2D + 1.2F + 1.6L

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

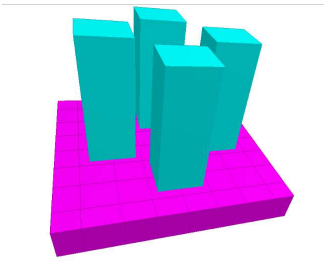
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:1.2D	7.76	bottom face	50	344	7.764	0.09	0.0002	71.8	619.2	12	182.72	175	646.6	9.12	11.4	0.088	tension control
105:0.9D	-6.87	top face	50	344	6.867	0.08	0.0002	63.5		12	182.72	175	646.6	9.12	11.4	0.088	tension control
105:0.9D	7.61	bottom face	50	344	7.607	0.09	0.0002	70.3		12	182.72	175	646.6	9.12	11.4	0.088	tension control
106:1.2D	-6.67	top face	50	344	6.674	0.08	0.0002	61.7		12	182.72	175	646.6	9.12	11.4	0.088	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.016	1	6.4	3.373	0.002	0.344	332.796	336.613	590.188	332.8	1257	> 6.4	ok	249.6	φVc>Vu	not required
	Min Qx	0.033	1	13.2	1.651			355.853			336.61	1260	> 13.2	ok	252.5	φVc>Vu	not required

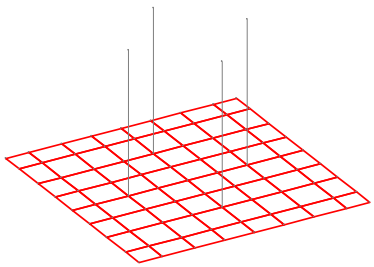
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106:1.2D	7.37	bottom face	50	344	7.374	0.08	0.0002	68.1	619.2	12	182.72	175	646.6	9.12	11.4	0.088	tension control
105:0.9D	-7.01	top face	50	344	7.005	0.08	0.0002	64.7		12	182.72	175	646.6	9.12	11.4	0.088	tension control
105:0.9D	7.17	bottom face	50	344	7.166	0.08	0.0002	66.2		12	182.72	175	646.6	9.12	11.4	0.088	tension control
106:1.2D	-6.83	top face	50	344	6.831	0.08	0.0002	63.1		12	182.72	175	646.6	9.12	11.4	0.088	tension control

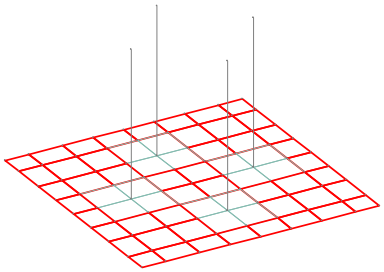
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.016	1	6.4	3.408	0.002	0.344	332.722	336.613	fc'590.2	332.72	350	> 6.4	ok	249.5	φVc>Vu	not required
	Min Qy	0.033	1	13.2	1.67			355.509			336.61	353	> 13.2	ok	252.5	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

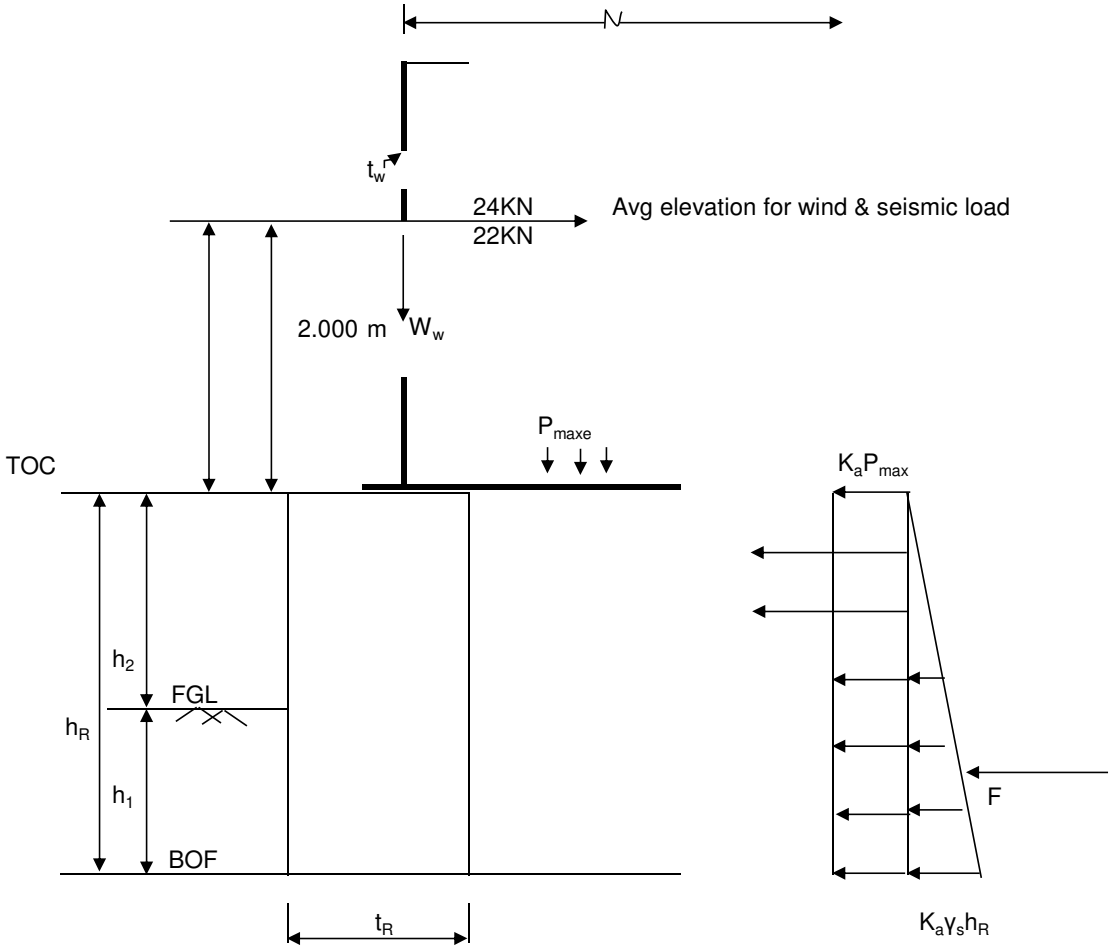
Along X Direction :
At Bottom 12ϕT 175 c/c
At Top 12ϕT 175 c/c
Along Z Direction :
At Bottom 12ϕT 175 c/c
At Top 12ϕT 175 c/c

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Design Calculation of Foundation for 3650-TNK-002



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	3.0	m
Self weight of tank =	35	KN
Self weight of tank per m of circumference,W_w =	3.71	KN/m
Maximum weight of Fluid =	450	KN
Pressure at base of tank ,P_{max}	63.7	KN/m^2
Design height of liquid, h_L =	3.5	m

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete,γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil,γ_s =	20	Kn/m^3
Allowable bearing pressure, q_{all}=	375	Kn/m^2
Coefficient of active earth pressure,K_a =	0.33	

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	1.8	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	0.6	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall,D_i =	2.55	m
Outer dia of ringwall,D_o =	3.45	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	2.83	m^3

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Check of soil bearing pressure (Max Fluid Load): For wind Load

Wind force = 22 kn acting at a height of 3.800 m from BOF

M_{OT} = Overturning moment = 83.6 KN-m

M_{OT} / S_{xx} = 29.6 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 83.3 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 113 KN/m^2 < 375.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 53.7 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 51.5 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 81.0 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 21.89 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 83.6 KN-m

M_{OR} = Restoring moment = (35+PI()*3*0.45*1.8*25)*3/2 = 338.8 KN-m

FOS = M_{OR} / M_{OT} 4.05 > 1.5 ok

Check for soil bearing pressure (Operating Fluid Load): For Seismic Load

Seismic force at center of mass equipment = 24 KN

acting at a height of 3.800 m from BOF

M_{OT} = Overturning moment = 91.2 KN-m

M_{OT} / S_{xx} = 32.2 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 83.3 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 106 KN/m^2 < 375.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 60.7 KN/m^2

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Check for differential settlement (Max Fluid Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + 0.7 \times P_{max} + h_R\gamma_c$ = 96.0 Kn/m^2 < 375.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 99.7 Kn/m^2 < 375.0 KN/m^2 Ok

As the tank encroaches the ring wall by 300mm out of total 450mm , the pressure under the tank is reduced to 300/450 = 0.7 times.

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 48.5 Kn/m

Hoop tension in ringwall, T = 1/2FD = 72.8 Kn

Factored design tension, T_u= 102 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 269 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1620 mm^2

$A_{s,reqd}$ (per face of ring wall)= 810 mm^2

Required : 12 dia bar @ 252 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1358 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 393 mm^2 ok

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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ATTACHMENT 1 INPUT DRAWINGS

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DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1015 | REV: A01

PREVAILING WIND DIRECTION | PLANT NORTH | 32° | GEODETIC NORTH | 125° | MECCA | W | S | E

3650-025 HOLD — 3650-PMP-008 CARBON TRANSFER WATER PUMP
3650-PMP-011 CARBON FINES FILTER FEED PUMP — 3650-028 HOLD
3650-PMP-103 POX CARBON AREA SUMP PUMP — 3650-072 HOLD
E=264212.413 N=2506103.562
TOS EL. +913.550
B.O.B.(+)913.590 (TYP)
B.O.B.(+)913.300
B.O.B.(+)913.100
TOC EL. +912.540
3650-THK-001 COMBINED CARBON SETTLER — 3650-038 HOLD
3650-PMP-012 CARBON FINES FILTER FEED PUMP — 3650-029 HOLD
3650-REA-001 CARBON ATTRITIONING TANK — 3650-032 HOLD
ID OF TANK ⌀ 5500
ID OF SETTLER ⌀ 6000
ID OF REACTOR ⌀ 1800
ID OF COLUMN ⌀ 1200
3650-TNK-006 CARBON PROCESS WATER TANK — 3650-044 HOLD
3650-PMP-007 CARBON TRANSFER WATER PUMP — 3650-024 HOLD
TOC EL. +912.700
3650-PMP-001 POX ELUTION PUMP — 3650-018 HOLD
3650-PMP-009 POX REGENERATED CARBON PUMP — 3650-026 HOLD
3610-TNK-001 POX CIL WASH SOLUTION TANK
3610-PMP-013 POX CIL WASH SOLUTION PUMP — 3610-016 HOLD
3610-PMP-012 POX CIL WASH SOLUTION PUMP — 3610-015 HOLD
3650-TNK-002 POX ELUTION TANK — 3650-040 HOLD
3650-PMP-010 POX REGENERATED CARBON PUMP — 3650-027 HOLD
3650-PMP-102 POX CARBON AREA SUMP PUMP — 3650-031 HOLD
3820-PMP-003 HYDROCHLORIC ACID DOSING PUMP — 3820-007 HOLD
3820-PMP-005 HYDROCHLORIC ACID DOSING PUMP — 3820-009 HOLD
3650-PMP-002 POX ELUTION PUMP — 3650-019 HOLD
TOS EL. +913.350
3650-TNK-001 POX ACID WASH TANK — 3650-039 HOLD
3650-PMP-006 POX ACID WASH PUMP — 3650-023 HOLD
TOC EL. +912.470
3650-COL-001 POX ACID WASH COLUMN — 3650-002 HOLD
3650-COL-002 POX ELUTION COLUMN — 3650-003 HOLD
3820-TNK-001 HYDROCHLORIC ACID STORAGE TANK — 3820-012 HOLD
3820-PMP-004 HYDROCHLORIC ACID DOSING PUMP — 3820-008 HOLD
TOS EL. +913.850
3650-PMP-101 POX ACID WASH AREA SUMP PUMP — 3650-030 HOLD
3650-PMP-005 POX ACID WASH PUMP — 3650-022 HOLD
3650-TNK-003 POX NEUTRALIZATION WASH TANK — 3650-041 HOLD
E=264197.660 N=2506067.163

4300 | 4000 | 3150 | 1050 | 660 | 4200 | 4200 | 5200 | 4200 | 5700 | 31800

3650/1 | 3650/2 | 3650/3 | 3650/4 | 3650/5 | 3650/6 | 3650/7 | 3650/8

3650/F | 3650/E | 3650/D | 3650/C | 3650/B | 3650/A

4550 | 6800 | 2850 | 3900 | 4000 | 4000

PLAN AT EL.(+)912.700

NOTE:
FOR LOAD TABLE REFER DRAWING NUMBER MD-216-3650-EG-ME-GAD-1030

0 1 10m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED. 05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +X, +Mx, +Y, +My

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m² — GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN.

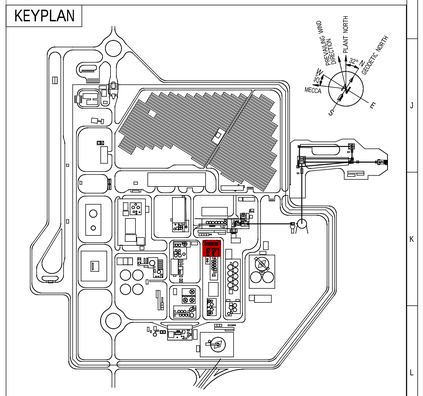
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.700 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3650-EG-ME-GAD-1030	LOAD TABLE
MD-216-3650-EG-ME-GAD-1018	LOAD PLAN, PLAN AT EL.+925.200 & EL.+927.400
MD-216-3650-EG-ME-GAD-1017	LOAD PLAN, PLAN AT EL.+918.700 & EL.+921.300
MD-216-3650-EG-ME-GAD-1016	LOAD PLAN, PLAN AT EL.+915.400
MD-216-3650-EG-ME-GAD-1002	GENERAL ARRANGEMENT DRAWING PLAN AT EL. +912.700

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	14-04-2020	ISSUED FOR APPROVAL	CK	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER	O19045_450_M_D_02975_001		REVISION 0
OUTOTEC DRAWING NUMBER	900707 _GAU01 _EE12 _MLD14 _36501	DOCUMENT ID OU100680658	REVISION 0
DRAWN	CK	14-04-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	14-04-2020	LOAD PLAN
ENGINEER	SC	14-04-2020	POX ACID WASH/ELUTION/CARBON REGENERATION
APPROVED	RL	14-04-2020	PLAN AT EL. +912.700
EMPLOYER			
SCALE 1:50	A0	DRAWING NUMBER MD-216-3650-EG-ME-GAD-1015	REV A01

DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1030 | REV: A01

LOADING POINT	P1.1/.../P1.4						P2						P3					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 2.5						- 12						- 12					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC							- 18						- 18					
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD	- 17																	
10 WIND ±X - DIRECTION		2.5		2					2					2				
11 WIND ±Y - DIRECTION		2.5			2					2					2			
12 EARTHQUAKE ±X - DIRECTION		1		1					2					2				
13 EARTHQUAKE ±Y - DIRECTION		1			1					2					2			
EQUIPMENT - NO.	3650-REA-001						3650-PMP-009						3650-PMP-010					
DESCRIPTION	CARBON ATTRITIONING TANK						POX REGENERATED CARBON PUMP						POX REGENERATED CARBON PUMP					

LOADING POINT	P4						P5.1/.../5.4						P6.1/.../6.4					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 12						- 10						- 10					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC	- 18																	
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD							- 50						- 50					
10 WIND ±X - DIRECTION			2				100	8					100	8				
11 WIND ±Y - DIRECTION				2			100		8				100		8			
12 EARTHQUAKE ±X - DIRECTION			2				57	4					57	4				
13 EARTHQUAKE ±Y - DIRECTION				2			57			4			57			4		
EQUIPMENT - NO.	3650-PMP-103						3650-COL-002						3650-COL-001					
DESCRIPTION	POX CARBON AREA SUMP PUMP						POX ELUTION COLUMN						POX ACID WASH COLUMN					

LOADING POINT	P7						P8						P9					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 25						- 12						- 25					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC							- 18											
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD	- 220												- 220					
10 WIND ±X - DIRECTION			18					2						18				
11 WIND ±Y - DIRECTION				18					2						18			
12 EARTHQUAKE ±X - DIRECTION			12					2						12				
13 EARTHQUAKE ±Y - DIRECTION				12					2						12			
EQUIPMENT - NO.	3650-TNK-003						3650-PMP-005						3650-TNK-001					
DESCRIPTION	POX NEUTRALIZATION WASH TANK						POX ACID WASH PUMP						POX ACID WASH TANK					

LOADING POINT	P10						P11						P12					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 12						- 12						- 50					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC	- 18						- 18											
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD													- 330					
10 WIND ±X - DIRECTION			2					2						-	14	-		
11 WIND ±Y - DIRECTION				2					2					-	-	14		
12 EARTHQUAKE ±X - DIRECTION			2					2						-	17	-		
13 EARTHQUAKE ±Y - DIRECTION				2					2					-	-	17		
EQUIPMENT - NO.	3650-PMP-006						3650-PMP-101						3820-TNK-001					
DESCRIPTION	POX ACID WASH PUMP						POX ACID WASH AREA SUMP PUMP						HYDROCHLORIC ACID STORAGE TANK					

LOADING POINT	P13						P14						P15					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 3						- 3						- 3					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC	- 5						- 5						- 5					
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD																		
10 WIND ±X - DIRECTION	-	0.5	-				-	0.5	-				-	0.5	-			
11 WIND ±Y - DIRECTION	-	-	0.5				-	-	0.5				-	-	0.5			
12 EARTHQUAKE ±X - DIRECTION	-	0.2	-				-	0.2	-				-	0.2	-			
13 EARTHQUAKE ±Y - DIRECTION	-	-	0.2				-	-	0.2				-	-	0.2			
EQUIPMENT - NO.	3820-PMP-004						3820-PMP-005						3820-PMP-003					
DESCRIPTION	HYDROCHLORIC ACID DOSING PUMP						HYDROCHLORIC ACID DOSING PUMP						HYDROCHLORIC ACID UNLOADING PUMP					

LOADING POINT	P16						P17.1/.../P17.3						P18					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 12						- 21						- 2					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING							- 130											
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC													- 4					
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD							- 170											
10 WIND ±X - DIRECTION			2					5						0.5				
11 WIND ±Y - DIRECTION				2					5						0.5			
12 EARTHQUAKE ±X - DIRECTION			2					3.5						0.1				
13 EARTHQUAKE ±Y - DIRECTION				2					3.5						0.1			
EQUIPMENT - NO.	3650-PMP-007						3610-TNK-001						3610-PMP-012					
DESCRIPTION	CARBON TRANSFER WATER PUMP						POX CIL TANK						POX CIL WASH SOLUTION PUMP					

LOADING POINT	P19						P20						P21					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 2						- 150						- 12					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC	- 4																	
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD							- 1500						- 18					
10 WIND ±X - DIRECTION		0.5						73						2				
11 WIND ±Y - DIRECTION			0.5						73						2			
12 EARTHQUAKE ±X - DIRECTION		0.1						83						2				
13 EARTHQUAKE ±Y - DIRECTION			0.1						83						2			
EQUIPMENT - NO.	3610-PMP-013						3650-TNK-006						3650-PMP-007					
DESCRIPTION	POX CIL WASH SOLUTION PUMP						CARBON PROCESS WATER TANK						CARBON TRANSFER WATER PUMP					

LOADING POINT	P22						P23.1/.../P23.8						P24					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 12						- 14						- 12					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC													- 18					
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD	- 18						- 170											
10 WIND ±X - DIRECTION			2					22	9.5					2				
11 WIND ±Y - DIRECTION				2			2	22		9.5					2			
12 EARTHQUAKE ±X - DIRECTION			2					25	11					2				
13 EARTHQUAKE ±Y - DIRECTION				2			2			11					2			
EQUIPMENT - NO.	3650-PMP-008						3650-THK-001						3650-PMP-012					
DESCRIPTION	CARBON TRANSFER WATER PUMP						COMBINED CARBON SETTLER						CARBON FINES FILTER FEED PUMP					

LOADING POINT	P25						P26						P27					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 12						- 35						- 12					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC	- 18												- 18					
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD							- 450											
10 WIND ±X - DIRECTION			2					22						2				
11 WIND ±Y - DIRECTION				2					22						2			
12 EARTHQUAKE ±X - DIRECTION			2					24						2				
13 EARTHQUAKE ±Y - DIRECTION				2					24						2			
EQUIPMENT - NO.	3650-PMP-011						3650-TNK-002						3650-PMP-002					
DESCRIPTION	CARBON FINES FILTER FEED PUMP						POX ELUTION TANK						POX ELUTION PUMP					

LOADING POINT	P28						P29.1/.../29.8						P30					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 12						- 30						- 10					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC	- 18																	
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD							- 35						- 15					
10 WIND ±X - DIRECTION			2															
11 WIND ±Y - DIRECTION				2														
12 EARTHQUAKE ±X - DIRECTION			2															
13 EARTHQUAKE ±Y - DIRECTION				2														
EQUIPMENT - NO.	3650-PMP-001						3650-FLT-001						3650-HEX-005					
DESCRIPTION	POX ELUTION PUMP						CARBON FINES FILTER						POX HEATER HEAT EXCHANGER					

LOADING POINT	P31						P32					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 10						- 10					
2 DEAD LOAD (SECONDARY EQUIPMENT)												
3 OPERATION LOAD 1. FILLING												
4 OPERATION LOAD 2. PRESSURE												
5 OPERATION LOAD 3. TEMPERATURE												
6 OPERATION LOAD 4. DYNAMIC												
7 OPERATION LOAD 5. OTHERS												
8 TEST LOAD FILLING / EMERGENCY												
9 LIVE LOAD	- 15						- 15					
10 WIND ±X - DIRECTION												
11 WIND ±Y - DIRECTION												
12 EARTHQUAKE ±X - DIRECTION												
13 EARTHQUAKE ±Y - DIRECTION												
EQUIPMENT - NO.	3650-HEX-004						3650-HEX-003					
DESCRIPTION	POX HEATER HEAT EXCHANGER						POX EVALUATE HEAT EXCHANGER					

LOADING POINT	P33						P34						P35					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 10						- 10						- 5					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC													- 8					
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD	- 15						- 15											
10 WIND ±X - DIRECTION										1					2			
11 WIND ±Y - DIRECTION											1					2		
12 EARTHQUAKE ±X - DIRECTION										0.3					1.5			
13 EARTHQUAKE ±Y - DIRECTION											0.3					1.5		
EQUIPMENT - NO.	3650-HEX-002						3650-HEX-001						3820-FAN-001					
DESCRIPTION	POX RECLAIM HEAT EXCHANGER						POX RECLAIM HEAT EXCHANGER						HYDROCHLORIC ACID AREA OFFGAS FAN					

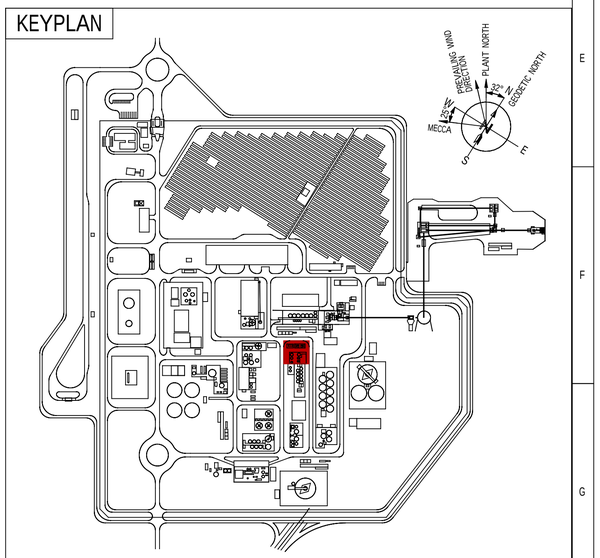
LOADING POINT	P36.1/.../P36.12						P37.1/.../37.4						P38.1/.../38.4					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 2						- 4						- 35					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC	- 3																	
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD							- 40						- 45					
10 WIND ±X - DIRECTION			2					14						12				
11 WIND ±Y - DIRECTION				1					14						12			
12 EARTHQUAKE ±X - DIRECTION			1.5					2						12				
13 EARTHQUAKE ±Y - DIRECTION				1.5					2						12			
EQUIPMENT - NO.	3650-SCR-001						3650-TNK-005						3650-FNC-001					
DESCRIPTION	POX CARBON DEWATERING SCREEN						POX QUENCH TANK						POX REGENERATION KILN					

LOADING POINT	P40.1/.../40.4						P41.1/.../41.4						P42					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 25						- 5						- 5					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC													- 8					
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD	- 45						- 10											
10 WIND ±X - DIRECTION			3					3						1				
11 WIND ±Y - DIRECTION				25					3						1			
12 EARTHQUAKE ±X - DIRECTION			2.5					2.5						0.3				
13 EARTHQUAKE ±Y - DIRECTION				12					2.5						0.3			
EQUIPMENT - NO.	3650-HOP-001						3650-SCR-002						3650-SCB-001					
DESCRIPTION	POX REGENERATION KILN FEED HOPPER						TRASH STATIC SCREEN						HYDROCHLORIC ACID AREA OFFGASDEMISTER					

LOADING POINT	P43.1/.../P43.2						P44.1/.../P44.2						P45.1/.../P45.2					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]			Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 9						- 5						- 7					
2 DEAD LOAD (SECONDARY EQUIPMENT)																		
3 OPERATION LOAD 1. FILLING																		
4 OPERATION LOAD 2. PRESSURE																		
5 OPERATION LOAD 3. TEMPERATURE																		
6 OPERATION LOAD 4. DYNAMIC																		
7 OPERATION LOAD 5. OTHERS																		
8 TEST LOAD FILLING / EMERGENCY																		
9 LIVE LOAD							- 10						- 18					
10 WIND ±X - DIRECTION			13															
11 WIND ±Y - DIRECTION				13														
12 EARTHQUAKE ±X - DIRECTION			9															
13 EARTHQUAKE ±Y - DIRECTION				9														
EQUIPMENT - NO.	3650-CHT-004						3650-FDB-001						3650-CHT-002					
DESCRIPTION	POX CARBON DE- WATERING SCREEN OVERSIZE CHUTE						POX CARBON DE- WATERING SCREEN FEED BOX						POX CARBON DE- WATERING SCREEN UNDERPAN					

LOADING POINT	P46.1/.../P46.2					
LOAD CASE	Vz[kN]	Hx[kN]		Hy[kN]		
1 DEAD LOAD EQUIPMENT	- 5					
2 DEAD LOAD (SECONDARY EQUIPMENT)						
3 OPERATION LOAD 1. FILLING						
4 OPERATION LOAD 2. PRESSURE						
5 OPERATION LOAD 3. TEMPERATURE						
6 OPERATION LOAD 4. DYNAMIC						
7 OPERATION LOAD 5. OTHERS						
8 TEST LOAD FILLING / EMERGENCY						
9 LIVE LOAD	- 10					
10 WIND ±X - DIRECTION						
11 WIND ±Y - DIRECTION						
12 EARTHQUAKE ±X - DIRECTION						
13 EARTHQUAKE ±Y - DIRECTION						
EQUIPMENT - NO.	3650-CHT-001					
DESCRIPTION	POX CARBON DE- WATERING SCREEN OVERSIZE CHUTE					

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3650-EG-ME-GAD-1018	LOAD PLAN, PLAN AT EL.+925.200 & EL.+927.400								
MD-216-3650-EG-ME-GAD-1017	LOAD PLAN, PLAN AT EL.+918.700 & EL.+921.300								
MD-216-3650-EG-ME-GAD-1016	LOAD PLAN, PLAN AT EL.+915.400								
MD-216-3650-EG-ME-GAD-1015	LOAD PLAN, PLAN AT EL.+912.700								
MD-216-3650-EG-ME-GAD-1002	GENERAL ARRANGEMENT DRAWING PLAN AT EL. +912.700	A01	14-04-2020	ISSUED FOR APPROVAL	NA	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_D_02918_059 | REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 _GAU01 _EE12 _MLD14 _36505 | DOCUMENT ID: OU100681424 | REVISION: 0

DRAWN	NA	14-04-2020
CHECKED	TV	14-04-2020
ENGINEER	SC	14-04-2020
APPROVED	RL	14-04-2020
EMPLOYER		
SCALE	-	A1

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
POX ACID WASH/ELUTION/CARBON REGENERATION LOAD TABLE

DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1030 | REV: A01

ATTACHMENT 2 STAAD INPUT - FDN OF 3650-REA-001

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18.05.2020
JOB NAME MMGP/3650-COL-001 & 002
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
CHECKER NAME SJM
APPROVED NAME AG
END JOB INFORMATION
UNIT METER KN
JOINT COORDINATES
7971 366.031 911 -382.65; 7973 366.031 911 -383.937;
7975 366.031 911 -382.168; 7976 366.031 911 -384.418;
7977 364.906 911 -383.293; 7978 367.156 911 -383.293;
7979 367.156 911 -382.168; 7980 367.156 911 -384.418;
7981 364.906 911 -382.168; 7982 364.906 911 -384.418;
7987 365.71 911 -384.418; 7988 366.353 911 -384.418;
7989 365.71 911 -382.168; 7990 366.353 911 -382.168;
7991 365.71 911 -383.293; 7992 366.031 911 -383.293;
7993 366.353 911 -383.293; 7994 365.71 911 -382.65;
7996 364.906 911 -382.65; 7997 366.353 911 -382.65;
7999 367.156 911 -382.65; 8000 365.71 911 -383.937;
8002 364.906 911 -383.937; 8003 366.353 911 -383.937;
8005 367.156 911 -383.937; 8006 364.906 911 -383.615;
8007 364.906 911 -382.971; 8008 367.156 911 -383.615;
8009 367.156 911 -382.971; 8011 366.031 911 -383.615;
8014 365.71 911 -382.971; 8015 366.031 911 -382.971;
8016 366.353 911 -382.971; 8017 365.71 911 -383.615;
8019 366.353 911 -383.615; 8052 366.698 911 -383.96;
8053 365.364 911 -383.96; 8054 366.698 911 -382.626;
8055 365.364 911 -382.626; 8056 366.698 912.75 -383.96;
8057 365.364 912.75 -383.96; 8058 366.698 912.75 -382.626;
8059 365.364 912.75 -382.626; 8060 365.364 911 -383.615;
8061 365.364 911 -383.293; 8062 365.364 911 -382.971;
8063 365.364 911 -382.168; 8065 365.364 911 -384.418;
8069 366.698 911 -384.418; 8070 366.698 911 -383.615;
8071 366.698 911 -383.293; 8072 366.698 911 -382.971;
8073 366.698 911 -382.168;
MEMBER INCIDENCES
4 8055 8059; 190 8052 8056; 193 8054 8058; 194 8053 8057;
ELEMENT INCIDENCES SHELL
281 7987 7976 7973 8000; 282 7976 7988 8003 7973;
287 8000 7973 8011 8017; 288 7973 8003 8019 8011;
293 8017 8011 7992 7991; 294 8011 8019 7993 7992;
299 7991 7992 8015 8014; 300 7992 7993 8016 8015;
305 8014 8015 7971 7994; 306 8015 8016 7997 7971;
311 7994 7971 7975 7989; 312 7971 7997 7990 7975;
371 7982 8065 8053 8002; 372 8065 7987 8000 8053;
373 8002 8053 8060 8006; 374 8053 8000 8017 8060;
375 8006 8060 8061 7977; 376 8060 8017 7991 8061;
377 7977 8061 8062 8007; 378 8061 7991 8014 8062;
379 8007 8062 8055 7996; 380 8062 8014 7994 8055;
381 7996 8055 8063 7981; 383 8055 7994 7989 8063;
413 7988 8069 8052 8003; 414 8069 7980 8005 8052;
415 8003 8052 8070 8019; 416 8052 8005 8008 8070;
417 8019 8070 8071 7993; 418 8070 8008 7978 8071;
419 7993 8071 8072 8016; 420 8071 7978 8009 8072;
421 8016 8072 8054 7997; 422 8072 8009 7999 8054;
423 7997 8054 8073 7990; 425 8054 7999 7979 8073;
START GROUP DEFINITION
ELEMENT
_FDN_SHEAR 281 282 287 288 293 294 299 300 305 306 311 312 375 TO 378 -
417 TO 420
_FDN_MOMENT 281 282 287 288 293 294 299 300 305 306 311 312 -
371 TO 381 383 413 TO 423 425
END GROUP DEFINITION
ELEMENT PROPERTY
281 282 287 288 293 294 299 300 305 306 311 312 371 TO 381 383 413 -
414 TO 423 425 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
```



```
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
4 190 193 194 PRIS YD 0.5 ZD 0.5
CONSTANTS
BETA 45 MEMB 190
BETA 135 MEMB 194
BETA 225 MEMB 4
BETA 315 MEMB 193
MATERIAL CONCRETE ALL
SUPPORTS
281 282 287 288 293 294 299 300 305 306 311 312 371 TO 381 383 413 -
414 TO 423 425 PLATE MAT DIRECT Y SUBGRADE 48000 PRINT
*************************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
281 282 287 288 293 294 299 300 305 306 311 312 371 TO 381 383 413 -
414 TO 423 425 PR GY -16
* LOAD DUE TO SURCHARGE
281 282 287 288 293 294 299 300 305 306 311 312 371 TO 381 383 413 -
414 TO 423 425 PR GY -20
JOINT LOAD
8056 TO 8059 FY -2.5
*************************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
8056 TO 8059 FY -17
*************************************************************************************************************
LOAD 3 LOADTYPE None  TITLE FLUID LOAD ( F )
JOINT LOAD
8056 TO 8059 FY -0.0001
*************************************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD ( W )
JOINT LOAD
8056 TO 8059 FX 2 FY -2.5 FZ 2
*************************************************************************************************************
LOAD 5 LOADTYPE None  TITLE SEISMIC LOAD ( E )
JOINT LOAD
8056 TO 8059 FX 1 FY -1 FZ 1
*************************************************************************************************************
* FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
*************************************************************************************************
LOAD COMB 101 1.4D + 1.4F
1 1.4 3 1.4
LOAD COMB 102 1.2D + 1.2F + 1.6L
1 1.2 3 1.2 2 1.6
LOAD COMB 103 1.2D + 1.0E + 0.5L
1 1.2 5 1.0 2 0.5
LOAD COMB 104 0.9D + 1.0E
1 0.9 5 1.0
LOAD COMB 105 0.9D + 1.6W
1 0.9 4 1.6
LOAD COMB 106 1.2D + 1.6W + 0.5L
1 1.2 4 1.6 2 0.5
*************************************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
LOAD COMB 201 D + F
1 1.0 3 1.0
LOAD COMB 202 D + F + L
1 1.0 2 1.0 3 1.0
LOAD COMB 203 D + F + W
1 1.0 3 1.0 4 1.0
LOAD COMB 204 0.6D + W
1 0.6 4 1.0
LOAD COMB 205 0.6D + 0.7E
1 0.6 5 0.7
LOAD COMB 206 D + 0.75W + 0.75L
1 1.0 4 0.75 2 0.75
*************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 106
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 4 190 193 194
CLB 0.05 MEMB 4 190 193 194
```

```
CLB 0.05 MEMB 4 190 193 194
CLS 0.05 ALL
CLT 0.05 MEMB 4 190 193 194
MAXMAIN 20 ALL
MINMAIN 20 ALL
MINSEC 10 ALL
DESIGN COLUMN 4 190 193 194
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18.05.2020
JOB NAME MMGP/3650-REA-001_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
CHECKER NAME SJM
APPROVED NAME AG
END JOB INFORMATION
UNIT METER KN
JOINT COORDINATES
8052 366.698 911 -383.96; 8053 365.364 911 -383.96;
8054 366.698 911 -382.626; 8055 365.364 911 -382.626;
8056 366.698 912.75 -383.96; 8057 365.364 912.75 -383.96;
8058 366.698 912.75 -382.626; 8059 365.364 912.75 -382.626;
MEMBER INCIDENCES
4 8055 8059; 190 8052 8056; 193 8054 8058; 194 8053 8057;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
4 190 193 194 PRIS YD 0.5 ZD 0.5
CONSTANTS
BETA 45 MEMB 190
BETA 135 MEMB 194
BETA 225 MEMB 4
BETA 315 MEMB 193
MATERIAL CONCRETE ALL
SUPPORTS
8052 TO 8055 FIXED
*********************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
8056 TO 8059 FY -2.5
*********************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
8056 TO 8059 FY -17
*********************************************************************************************************
LOAD 3 LOADTYPE None  TITLE FLUID LOAD ( F )
JOINT LOAD
8056 TO 8059 FY -0.0001
*********************************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD ( W )
JOINT LOAD
8056 TO 8059 FX 2 FY -2.5 FZ 2
*********************************************************************************************************
LOAD 5 LOADTYPE None  TITLE SEISMIC LOAD ( E )
JOINT LOAD
8056 TO 8059 FX 1 FY -1 FZ 1
*********************************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE & STABILITY CHECK
LOAD COMB 201 D + F
1 1.0 3 1.0
LOAD COMB 202 D + F + L
1 1.0 2 1.0 3 1.0
LOAD COMB 203 D + F + W
1 1.0 3 1.0 4 1.0
LOAD COMB 204 0.6D + W
1 0.6 4 1.0
LOAD COMB 205 0.6D + 0.7E
1 0.6 5 0.7
LOAD COMB 206 D + 0.75W + 0.75L
1 1.0 4 0.75 2 0.75
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 3 STAAD OUTPUT - FDN OF 3650-REA-001

PROPRIETARY & CONFIDENTIAL

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```
          **************************************************
          *                                                *
          *          STAAD.Pro V8i SELECTseries5           *
          *          Version  20.07.10.64                  *
          *          Proprietary Program of                *
          *          Bentley Systems, Inc.                 *
          *          Date=    MAY 26, 2020                 *
          *          Time=    20: 3:11                     *
          *                                                *
          *     USER ID: LNT ECC DIV                       *
          **************************************************
```

```
    1. STAAD SPACE
INPUT FILE: 3650-REA-001.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 18.05.2020
    4. JOB NAME MMGP/3650-COL-001 _002
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. CHECKER NAME SJM
   10. APPROVED NAME AG
   11. END JOB INFORMATION
   12. UNIT METER KN
   13. JOINT COORDINATES
   14. 7971 366.031 911 -382.65; 7973 366.031 911 -383.937
   15. 7975 366.031 911 -382.168; 7976 366.031 911 -384.418
   16. 7977 364.906 911 -383.293; 7978 367.156 911 -383.293
   17. 7979 367.156 911 -382.168; 7980 367.156 911 -384.418
   18. 7981 364.906 911 -382.168; 7982 364.906 911 -384.418
   19. 7987 365.71 911 -384.418; 7988 366.353 911 -384.418
   20. 7989 365.71 911 -382.168; 7990 366.353 911 -382.168
   21. 7991 365.71 911 -383.293; 7992 366.031 911 -383.293
   22. 7993 366.353 911 -383.293; 7994 365.71 911 -382.65
   23. 7996 364.906 911 -382.65; 7997 366.353 911 -382.65
   24. 7999 367.156 911 -382.65; 8000 365.71 911 -383.937
   25. 8002 364.906 911 -383.937; 8003 366.353 911 -383.937
   26. 8005 367.156 911 -383.937; 8006 364.906 911 -383.615
   27. 8007 364.906 911 -382.971; 8008 367.156 911 -383.615
   28. 8009 367.156 911 -382.971; 8011 366.031 911 -383.615
   29. 8014 365.71 911 -382.971; 8015 366.031 911 -382.971
   30. 8016 366.353 911 -382.971; 8017 365.71 911 -383.615
   31. 8019 366.353 911 -383.615; 8052 366.698 911 -383.96
   32. 8053 365.364 911 -383.96; 8054 366.698 911 -382.626
   33. 8055 365.364 911 -382.626; 8056 366.698 912.75 -383.96
   34. 8057 365.364 912.75 -383.96; 8058 366.698 912.75 -382.626
   35. 8059 365.364 912.75 -382.626; 8060 365.364 911 -383.615
   36. 8061 365.364 911 -383.293; 8062 365.364 911 -382.971
   37. 8063 365.364 911 -382.168; 8065 365.364 911 -384.418
   38. 8069 366.698 911 -384.418; 8070 366.698 911 -383.615
```



```
   39. 8071 366.698 911 -383.293; 8072 366.698 911 -382.971
   40. 8073 366.698 911 -382.168
   41. MEMBER INCIDENCES
   42. 4 8055 8059; 190 8052 8056; 193 8054 8058; 194 8053 8057
   43. ELEMENT INCIDENCES SHELL
   44. 281 7987 7976 7973 8000; 282 7976 7988 8003 7973
   45. 287 8000 7973 8011 8017; 288 7973 8003 8019 8011
   46. 293 8017 8011 7992 7991; 294 8011 8019 7993 7992
   47. 299 7991 7992 8015 8014; 300 7992 7993 8016 8015
   48. 305 8014 8015 7971 7994; 306 8015 8016 7997 7971
   49. 311 7994 7971 7975 7989; 312 7971 7997 7990 7975
   50. 371 7982 8065 8053 8002; 372 8065 7987 8000 8053
   51. 373 8002 8053 8060 8006; 374 8053 8000 8017 8060
   52. 375 8006 8060 8061 7977; 376 8060 8017 7991 8061
   53. 377 7977 8061 8062 8007; 378 8061 7991 8014 8062
   54. 379 8007 8062 8055 7996; 380 8062 8014 7994 8055
   55. 381 7996 8055 8063 7981; 383 8055 7994 7989 8063
   56. 413 7988 8069 8052 8003; 414 8069 7980 8005 8052
   57. 415 8003 8052 8070 8019; 416 8052 8005 8008 8070
   58. 417 8019 8070 8071 7993; 418 8070 8008 7978 8071
   59. 419 7993 8071 8072 8016; 420 8071 7978 8009 8072
   60. 421 8016 8072 8054 7997; 422 8072 8009 7999 8054
   61. 423 7997 8054 8073 7990; 425 8054 7999 7979 8073
   62. START GROUP DEFINITION
   63. ELEMENT
   64. _FDN_SHEAR 281 282 287 288 293 294 299 300 305 306 311 312 375 TO 378 -
   65. 417 TO 420
   66. _FDN_MOMENT 281 282 287 288 293 294 299 300 305 306 311 312 -
   67. 371 TO 381 383 413 TO 423 425
   68. END GROUP DEFINITION
   69. ELEMENT PROPERTY
   70. 281 282 287 288 293 294 299 300 305 306 311 312 371 TO 381 383 413 -
   71. 414 TO 423 425 THICKNESS 0.4
   72. DEFINE MATERIAL START
   73. ISOTROPIC CONCRETE
   74. E 2.17185E+007
   75. POISSON 0.17
   76. DENSITY 25
   77. ALPHA 1E-005
   78. DAMP 0.05
   79. TYPE CONCRETE
   80. STRENGTH FCU 27579
   81. END DEFINE MATERIAL
   82. MEMBER PROPERTY AMERICAN
   83. 4 190 193 194 PRIS YD 0.5 ZD 0.5
   84. CONSTANTS
   85. BETA 45 MEMB 190
   86. BETA 135 MEMB 194
   87. BETA 225 MEMB 4
   88. BETA 315 MEMB 193
   89. MATERIAL CONCRETE ALL
   90. SUPPORTS
   91. 281 282 287 288 293 294 299 300 305 306 311 312 371 TO 381 383 413 -
   92. 414 TO 423 425 PLATE MAT DIRECT Y SUBGRADE 48000 PRINT
```

JOINT	INFLUENCE AREA
7971	0.129085
7973	0.129085
7975	0.077484
7976	0.077324
7977	0.073737
7978	0.073737
7979	0.054501
7980	0.054417
7981	0.054501
7982	0.054417
7987	0.079711
7988	0.079716
7989	0.079855
7990	0.079859
7991	0.107387
7992	0.103526
7993	0.107391
7994	0.133900
7996	0.091942
7997	0.133905
7999	0.091942
8000	0.133900
8002	0.091942
8003	0.133905
8005	0.091942
8006	0.074396
8007	0.074308
8008	0.074396
8009	0.074308
8011	0.103526
8014	0.107737
8015	0.103362
8016	0.107740
8017	0.107885
8019	0.107887
8052	0.161203
8053	0.161402
8054	0.161203
8055	0.161402
8060	0.132910
8061	0.129444
8062	0.132859
8063	0.093263
8065	0.093217
8069	0.093101
8070	0.132746
8071	0.129284
8072	0.132695
8073	0.093148
TOTAL	5.062539

```
93. ********************************************************************************
94. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
```



```
 95. SELFWEIGHT Y -1
 96. ELEMENT LOAD
 97. * LOAD DUE TO BACKFILL
 98. 281 282 287 288 293 294 299 300 305 306 311 312 371 TO 381 383 413 -
 99. 414 TO 423 425 PR GY -16
100. * LOAD DUE TO SURCHARGE
101. 281 282 287 288 293 294 299 300 305 306 311 312 371 TO 381 383 413 -
102. 414 TO 423 425 PR GY -20
103. JOINT LOAD
104. 8056 TO 8059 FY -2.5
105. ********************************************************************************
106. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
107. JOINT LOAD
108. 8056 TO 8059 FY -17
109. ********************************************************************************
110. LOAD 3 LOADTYPE NONE  TITLE FLUID LOAD ( F )
111. JOINT LOAD
112. 8056 TO 8059 FY -0.0001
113. ********************************************************************************
114. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD ( W )
115. JOINT LOAD
116. 8056 TO 8059 FX 2 FY -2.5 FZ 2
117. ********************************************************************************
118. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD ( E )
119. JOINT LOAD
120. 8056 TO 8059 FX 1 FY -1 FZ 1
121. ********************************************************************************
122. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
123. ********************************************************************************
124. LOAD COMB 101 1.4D + 1.4F
125. 1 1.4 3 1.4
126. LOAD COMB 102 1.2D + 1.2F + 1.6L
127. 1 1.2 3 1.2 2 1.6
128. LOAD COMB 103 1.2D + 1.0E + 0.5L
129. 1 1.2 5 1.0 2 0.5
130. LOAD COMB 104 0.9D + 1.0E
131. 1 0.9 5 1.0
132. LOAD COMB 105 0.9D + 1.6W
133. 1 0.9 4 1.6
134. LOAD COMB 106 1.2D + 1.6W + 0.5L
135. 1 1.2 4 1.6 2 0.5
136. ********************************************************************************
137. * UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
138. LOAD COMB 201 D + F
139. 1 1.0 3 1.0
140. LOAD COMB 202 D + F + L
141. 1 1.0 2 1.0 3 1.0
142. LOAD COMB 203 D + F + W
143. 1 1.0 3 1.0 4 1.0
144. LOAD COMB 204 0.6D + W
145. 1 0.6 4 1.0
146. LOAD COMB 205 0.6D + 0.7E
147. 1 0.6 5 0.7
148. LOAD COMB 206 D + 0.75W + 0.75L
149. 1 1.0 4 0.75 2 0.75
150. ********************************************************************************
```

151. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 53 NUMBER OF MEMBERS 4
NUMBER OF PLATES 36 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 49

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 46/ 13/ 45 DOF
TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 171
TOTAL LOAD COMBINATION CASES = 12 SO FAR.
SIZE OF STIFFNESS MATRIX = 8 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.3/ 146166.1 MB

152. LOAD LIST 101 TO 106
153. START CONCRETE DESIGN

154. CODE ACI
155. TRACK 2 ALL
156. FC 35000 MEMB 4 190 193 194
157. CLB 0.05 MEMB 4 190 193 194
158. CLS 0.05 ALL
159. CLT 0.05 MEMB 4 190 193 194
160. MAXMAIN 20 ALL
161. MINMAIN 20 ALL
162. MINSEC 10 ALL
163. DESIGN COLUMN 4 190 193 194

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

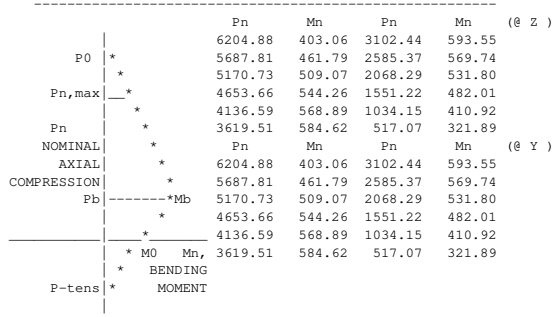
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3052.41	594.14	194.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
222.73	-1039.71	5.38	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3052.41	594.14	194.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
222.73	-1039.71	5.38	0.00	0.00000



COLUMN NO. 190 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

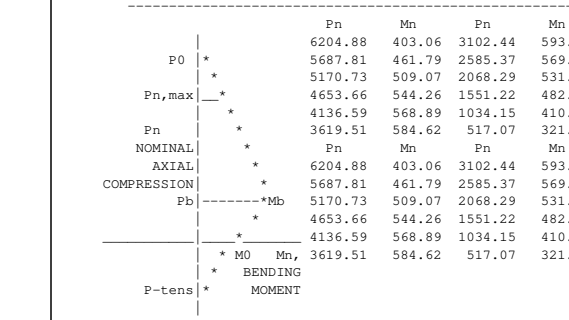
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3052.41	594.14	194.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
222.73	-1039.71	5.38	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3052.41	594.14	194.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
222.73	-1039.71	5.38	0.00	0.00000



COLUMN NO. 193 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

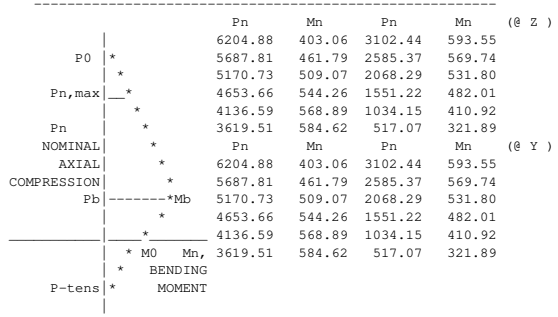
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3052.41	594.14	194.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
222.73	-1039.71	5.38	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3052.41	594.14	194.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
222.73	-1039.71	5.38	0.00	0.00000



COLUMN NO. 194 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.005	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3052.41	594.14	194.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
222.73	-1039.71	5.38	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8402.44	6721.95	3052.41	594.14	194.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
222.73	-1039.71	5.38	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

164. END CONCRETE DESIGN
165. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 26,2020 TIME= 20: 3:17 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

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            *****************************************************
            *                                                   *
            *           STAAD.Pro V8i SELECTseries5             *
            *           Version  20.07.10.64                    *
            *           Proprietary Program of                  *
            *           Bentley Systems, Inc.                   *
            *           Date=    MAY 26, 2020                   *
            *           Time=    20: 4:52                       *
            *                                                   *
            *      USER ID: LNT ECC DIV                         *
            *****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: 3650-REA-001_STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 18.05.2020
     4. JOB NAME MMGP/3650-REA-001_STABILITY CHECK
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. CHECKER NAME SJM
    10. APPROVED NAME AG
    11. END JOB INFORMATION
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 8052 366.698 911 -383.96; 8053 365.364 911 -383.96
    15. 8054 366.698 911 -382.626; 8055 365.364 911 -382.626
    16. 8056 366.698 912.75 -383.96; 8057 365.364 912.75 -383.96
    17. 8058 366.698 912.75 -382.626; 8059 365.364 912.75 -382.626
    18. MEMBER INCIDENCES
    19. 4 8055 8059; 190 8052 8056; 193 8054 8058; 194 8053 8057
    20. DEFINE MATERIAL START
    21. ISOTROPIC CONCRETE
    22. E 2.17185E+007
    23. POISSON 0.17
    24. DENSITY 25
    25. ALPHA 1E-005
    26. DAMP 0.05
    27. TYPE CONCRETE
    28. STRENGTH FCU 27579
    29. END DEFINE MATERIAL
    30. MEMBER PROPERTY AMERICAN
    31. 4 190 193 194 PRIS YD 0.5 ZD 0.5
    32. CONSTANTS
    33. BETA 45 MEMB 190
    34. BETA 135 MEMB 194
    35. BETA 225 MEMB 4
    36. BETA 315 MEMB 193
    37. MATERIAL CONCRETE ALL
    38. SUPPORTS
```



```
    39. 8052 TO 8055 FIXED
    40. ********************************************************************************
    41. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
          MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    42. SELFWEIGHT Y -1
    43. JOINT LOAD
    44. 8056 TO 8059 FY -2.5
    45. ********************************************************************************
    46. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
    47. JOINT LOAD
    48. 8056 TO 8059 FY -17
    49. ********************************************************************************
    50. LOAD 3 LOADTYPE NONE  TITLE FLUID LOAD ( F )
    51. JOINT LOAD
    52. 8056 TO 8059 FY -0.0001
    53. ********************************************************************************
    54. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD ( W )
    55. JOINT LOAD
    56. 8056 TO 8059 FX 2 FY -2.5 FZ 2
    57. ********************************************************************************
    58. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD ( E )
    59. JOINT LOAD
    60. 8056 TO 8059 FX 1 FY -1 FZ 1
    61. ********************************************************************************
    62. * UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE _STABILITY CHECK
    63. LOAD COMB 201 D + F
    64. 1 1.0 3 1.0
    65. LOAD COMB 202 D + F + L
    66. 1 1.0 2 1.0 3 1.0
    67. LOAD COMB 203 D + F + W
    68. 1 1.0 3 1.0 4 1.0
    69. LOAD COMB 204 0.6D + W
    70. 1 0.6 4 1.0
    71. LOAD COMB 205 0.6D + 0.7E
    72. 1 0.6 5 0.7
    73. LOAD COMB 206 D + 0.75W + 0.75L
    74. 1 1.0 4 0.75 2 0.75
    75. ********************************************************************************
    76. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**8**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 4/ 4/ 6 DOF
TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 6 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 146165.6 MB

77. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 26,2020 TIME= 20: 4:56 ****

****
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* http://selectservices.bentley.com/en-US/ *
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